

Article

Assessing the Sustainability of Local Construction Materials in Ecuador Through an Interdisciplinary Approach

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Abstract: Worldwide, concrete is used as the material of choice in the construction sector. In Latin American countries, concrete is primarily used in structural elements, because it is a low-maintenance, versatile, and economical material. With its characteristics, concrete has made it possible to cover Ecuador's housing deficit to some extent. However, this construction material does not entirely respond to the needs of local communities, who are especially concerned about the displacement of traditional techniques and the loss of cultural heritage. In Ecuador, different fiber- and earth-based materials have been used in the construction sector for a long time. These materials present different technical, environmental, social, and cultural characteristics. This study aimed to evaluate the sustainability of the production of six local materials in Ecuador under three dimensions: environmental, social, and cultural, using local data. The data were collected in local production workshops, and life-cycle-based methodologies were applied. The results showed the relevance of developing regional inventories to represent the reality of production within the territory. Additionally, integrating the evaluation of environmental, social, and cultural dimensions shed light on the current situation of the building materials production sector, identifying possible trade-offs or synergies with a sustainable and interdisciplinary approach.

Keywords: sustainable construction; local materials; life cycle thinking; environmental impacts; social LCA; cultural heritage



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1. Introduction

The construction sector significantly contributes to GDP worldwide [1,2], with Ecuador's share at 6.4% in 2022 [3]. However, rapid urban growth and increased demand for materials raise sustainability concerns. Globally, construction consumes 40% of natural resources and 36% of energy, generating 40% of waste and 33% of greenhouse gas emissions [4,5]. In Ecuador, industrialization of the construction sector has led to 48% natural resource consumption, 7% energy use, and 15% fossil fuel within the country [6]. Building materials play a crucial role, being responsible for 82% of carbon emissions from

buildings; in addition, 38% of the total energy used by buildings during their whole lifetime comes from the energy embodied in construction materials, and therefore associated with their own value chain [7] and production process.

In Ecuador, as in other Latin American countries, concrete dominates construction, due to its low maintenance, versatility, and cost-effectiveness [8]. While it has helped address the housing deficit, its environmental sustainability is a concern [9]. On this line, the environmental awareness pushed by The United Nations Agenda for Sustainable Development [10] has driven interest in vernacular materials and traditional techniques, which require less transport and have a lower embodied energy [11–15]. However, research on these materials is limited, focusing mainly on Europe and Asia [16], as seen in sustainability databases like ecoinvent [17]. To highlight the lack of local data at the Latin American and Ecuadorian levels, we can compare the availability of datasets with other regions in the ecoinvent database [17]. For instance, ecoinvent includes 2272 datasets for European regions, which provide an averaged representation of Europe. In contrast, only 31 datasets are available for the Latin American and Caribbean region.

While concrete is the dominant material in Ecuador, due to its affordability and durability, its environmental impact drives interest in vernacular materials. These materials, often locally sourced and less resource-intensive, can reduce environmental footprints. However, data on their sustainability in Latin America remain scarce. Ecuador has only 65 datasets of construction materials in the ecoinvent database (10 of these datasets are the result of the present research), compared to Colombia (140) and Peru (177), highlighting a lack of local impact assessments. Using foreign datasets can lead to a 19% variation in environmental impact estimations [18]. Making it difficult to understand the real environmental impacts of the construction sector in Ecuador.

From a socio-economic perspective, the construction sector has a high proportion of informal workers, with 77% in Ecuador [19]. This raises issues of occupational health, fair wages, working hours, and social security. Discussions on sustainability often overlook these aspects and focus mainly on the environmental dimension. Few studies have assessed the social impacts of material life cycles in construction, covering limited materials (e.g., concrete, steel, bamboo) [20,21] and indicators (e.g., job creation) [21]. In addition, local communities are concerned about industrialized and imported materials, which meet housing needs but do not meet cultural and functional needs [22,23].

Therefore, we aimed to assess the impacts of the production of six local traditional and contemporary construction materials in Ecuador through an interdisciplinary approach, thus understanding the complexity of a reality composed of multiple dimensions (environmental, social, and cultural), their relationships, and possible trade-offs or synergies [24]. The analyzed materials were bamboo, adobe brick, clay brick (both molded and extruded), clay roof tile, and concrete block. Through this approach, we addressed sustainability as a complex system [25] in the construction sector, where problems and solutions go beyond the scope of a single discipline [26], since their effects are simultaneous [27] in the dimensions that make up the system. This makes it a crucial issue to be addressed. While the economic dimension falls outside the scope of this research, this does not diminish its significance. Instead, our study prioritized an in-depth exploration of the social dimension and introduced an initial examination of the cultural dimension. In this way, the application of life cycle assessment (LCA) in both environmental and social dimensions, alongside the ethnographic method, enabled a holistic approach to challenges in the construction sector, particularly in material production.

Environmental LCA (E-LCA) helped evaluate materials and identify their environmental impact, allowing for the selection of local alternatives with lower impacts than concrete and reflecting real production conditions in Ecuador by generating datasets. Social LCA

(S-LCA) analyzed social impacts and provides insight into the living conditions of workers. The results of both methodologies will guide sustainable decisions in the construction sector in Ecuador. The ethnographic method helped to identify culturally significant materials and their relation to the identity of those involved in their production, revealing their value beyond the economic. Finally, this research goes beyond isolated methodological results, aiming for an interdisciplinary sustainability assessment. It examined building material production as a complex system, where environmental, social, and cultural factors interact dynamically, identifying materials with high sustainability potential.

The article is structured as follows. Section 2 details the methodologies used for the selection of the local construction materials in Ecuador (Section 2.1), the environmental and social assessment (Section 2.2), and the identification of cultural characteristics and meanings (Section 2.3). Section 3 shows the results of the selection of the local construction materials, the results of both environmental and social life cycle assessments, and the results of the identification of cultural meanings. Section 4 reports a discussion based on the integration of the results of Section 3, and finally, Section 5 draws conclusions and provides an outlook for future research.

2. Materials and Methods

The sustainability assessment of the production of local construction materials in Ecuador involved three main steps and methods:

1. Selection of five local construction materials, beside concrete, through a multi-criteria method
2. Assessment of environmental and social sustainability through the application of life cycle analysis methodologies
3. Identification of cultural meanings associated with the production of the selected materials through the ethnographic method.

2.1. Selection of Local Construction Materials

A multi-criteria analysis was applied to rank alternatives according to the weights assigned to certain criteria [28] to select at least five locally produced materials in Ecuador.

The preliminary list of local materials was elaborated according to the knowledge gathered by the authors through a literature analysis of the materials produced in the whole country and how much they are used in the construction sector [29]. The materials were grouped into traditional materials based on natural fibers, traditional materials based on earth, and contemporary materials beside concrete blocks. The next step in the application of the multi-criteria methodology was to define the criteria and subcriteria. These were selected based on the following considerations: the objectives of the research project, especially in relation to the four dimensions of sustainability evaluated, and the practical feasibility of collecting useful data on the materials that would allow us to correctly apply the three methodologies used in the study.

For the stage of assigning weights to the criteria and subcriteria, the logic of the Delphi method [30] was followed, with the aim of constructing a table of weights for the selection of building materials. In the first round, the weights were assigned based on the contributions of the research team and their expertise in the architectural, environmental, and sociological fields. The second round involved the participation of three experts, who were given the proposed weights generated in the first round to review, provided feedback, and adjusted the weights based on their knowledge and experience. At the end of the two rounds, the research team reviewed the final version of the assigned weights (Table 1), which allowed a consensus to be reached on the assignment of weights, to move to the materials selection stage.

Table 1. Criteria and subcriteria for the selection of materials, with associated weights based on own elaboration.

Criteria	Weight	Subcriteria	Weight
Technical	30%	Material use in heritage buildings	20%
		Material use in modern buildings	20%
		Data availability and accessibility	25%
		Thermal conductivity	10%
		Wall or roofing materials	25%
Environmental	35%	Waste generated in production	20%
		Use of local raw materials	20%
		Water consumption	20%
		Recyclable or reusable potential	20%
		Embedded energy	20%
Social and cultural	35%	Use of traditional/artisanal production techniques	30%
		Local cultural value	30%
		Vernacular material	25%
		Historical use of the material in the area	15%

With the list of preliminary materials and the criteria and subcriteria, the materials were evaluated using a scale from 1 to 3. This allowed us to obtain the final scores for each of the materials and the final list of materials to be evaluated based on the following specifications. The highest-scoring material within the group of natural fibers, the two highest-scoring materials within the group of traditional earth-based materials, and the three highest-scoring materials within the group of contemporary materials were selected. With these considerations, the final list of selected materials was composed as follows. For traditional materials based on natural fibers, bamboo (Spanish name: caña guadua) was selected. Three types of bamboo will be presented in the results, bamboo culm, bamboo pole, and flattened bamboo, which are part of the same production process, but depending on consumer needs can be sold at different stages, so they are presented separately. For traditional materials based on earth, adobe and molded clay brick (Spanish name: ladrillo panelón) were selected.

For contemporary materials, extruded clay brick (Spanish name: ladrillo tochano), clay roof tile, and concrete block (cement-based material, which is why cement was not selected as another material for the study) were selected. Regarding clay roof tiles, an additional production step warranted the analysis of two distinct types: standard clay roof tiles and glazed clay roof tiles, the latter are treated with lead oxide and silica for glazing.

Since the chosen materials are available in different provinces of the country, for the data collection of the production processes, the team selected the provinces where such materials are most produced. Two provinces and their respective capitals were identified:

- For bamboo, the province of Manabi with its capital, Portoviejo, was selected [31].
- For adobe, clay bricks, and tiles, the province of Azuay with its capital, Cuenca, was selected [32–34].
- For concrete blocks, due its presence all over the country [8], it was not necessary to choose a specific location. This widespread availability ensures that this material can be sourced from various locations, as per current project needs.

For more information on the production processes and technical characteristics of the selected materials, please refer to Supplementary Files S1 and S2.

2.2. Environmental and Social Assessment Through Life-Cycle-Based Methodologies

The environmental and social impacts of the selected materials' production processes were assessed using life-cycle methodologies [35,36]. The data for each material's life cycle inventory (LCI), environmental and social, were collected from thirteen production workshops through direct observation, interviews, and surveys of the producers. These workshops were selected because they were the most qualitatively representative of the area in which the production of each of the materials is developed. This was based on the antiquity of some of the workshops and the fact that several of the selected workshops were able to mechanize the production of certain contemporary materials. In addition, the production stages within each workshop were similar to each other in field visits and preliminary interviews with the producers. These processes were also contrasted with those mentioned in the literature, to verify their coincidence. The information for the selection of the workshops was obtained through preliminary visits to the study areas, conversations with members of the community, and recommendations from experts.

For the environmental life cycle assessment (E-LCA), the objective was to determine the environmental impacts associated with producing 1 kg of each of the selected local materials in Ecuador. Since the study focuses on the production processes of the materials, comparisons between materials will only be possible at the production level. At the level of use of the materials, i.e., in the construction of a house, the results do not allow a comparison between materials, since the function of 1 kg of, for example, bamboo is different from 1 kg of a brick when used for the construction of a house, but this evaluation is the first step in allowing the evaluation of a house built with the analyzed materials. The system boundaries were established as cradle-to-gate, i.e., from the extraction of raw materials which enter the factory to the production and storage of the final product (i.e., the material ready to be used in buildings).

To develop the environmental Life Cycle Inventory (LCI), the quantities of energy, raw materials, and chemicals were calculated and documented as input flows, while the quantities of the reference product, by-products, emissions, and solid residues were recorded as output flows. Additionally, on-site measurements were conducted in the workshops, to obtain the necessary volume and mass data. This data collection process was carried out using standardized sheets that tracked inputs (such as raw materials, water, energy, and others) and outputs (including waste, gases/vapors, and effluents) for each stage of the material production process. The LCI of the production of each material was elaborated according to the guidelines established by ecoinvent [37–39], the most used background life cycle database for life-cycle-related data, with the motivation that these datasets have a format that can be published and used at a global level. For the background data, ecoinvent v.3.9.1 was used.

Two methods available in SimaPro software v. 9.4.0.2 were then used to calculate the environmental impact, the IPCC 2021 [40], and the Environmental Footprint (EF 3.0) [41]. These methods were selected due to their international recognition and standardized frameworks, which facilitate the comparison of environmental impacts across different studies and materials. In addition, they generate results that complement each other, thus the CO₂e calculation obtained with the IPCC method compliments the results of the EF method that considers additional aspects such as water use, soil degradation, environmental toxicity, and consumption of natural resources.

The EF method includes 16 impact categories (climate change, ozone depletion, ionizing radiation, photochemical ozone formation, particulate matter, human toxicity, non-

cancer, human toxicity, cancer, acidification, eutrophication, freshwater, eutrophication, marine, eutrophication, terrestrial, ecotoxicity, freshwater, land use, water use, resource use, fossils, resource use, minerals, and metals) and provides normalized and weighted results. Normalization and weighting steps were used to facilitate the summation of the various environmental impacts and obtain the individual contribution of each category to the overall impact, expressed in milipoints (mPt) per kg of material produced [41,42]. The IPCC 2021 method offers an updated approach to assess climate change compared to the one available in EF v.3.0 (based on IPCC 2013) [43], allowing pointing out the differences between artisanal and mechanized or semi-mechanized production processes, while the EF v.3.0 method addresses multiple impact categories based on different indicators and mathematical models, also allowing us to make a comparison with the LCI results of producing the same/similar materials in other regions of the world, by using the datasets already available in ecoinvent v.3.9.1.

For the social life cycle assessment (S-LCA), it was defined as an objective and scope to determine the social performance of the artisanal and semi-mechanized production processes of building materials in the workshops. This analysis follows a 'door-to-door' approach, focusing on a single stakeholder: workers. In the case of this study, the stakeholder-workers were the producers.

The system and its boundaries were defined according to the characteristics of the production processes of the selected materials. Since most of them share a similar structure, except for the bamboo process, the boundary of the system started with the arrival of the raw material at the workshop and ended when the building material is finished and stored ready to be sold. In the case of bamboo, the transfer from the bamboo plantation to the preservation center, where the final processes are carried out and where the ready material is stored, was taken into account.

To create the social LCI, as recommended in the Guidelines for Social Life Cycle Assessment of Products and Organizations [36], the primary data were collected at a local level through direct interviews, with open questions and surveys applied to workers (the producers of the selected materials). The interview scripts and survey questionnaires were structured according to the indicators selected for the social assessment. The importance of collecting local data in the case of S-LCA lies in the specificity of each system evaluated and the context in which the system is developed. Therefore, this research provides an experience of applying the SLCA methodology to the production of building materials and a set of contextualized indicators that could be a starting point for applications and testing in other countries.

It is important to address data quality and completeness, as this is critical to ensure the reliability and validity of results and to reach useful conclusions. Although there is no single method for assessing data quality, as data may be subject to different perspectives, the S-LCA guidance [36] suggests defining relevant aspects of data quality, such as the timeliness and geographical or technical conformity of datasets. For this reason, a pedigree matrix (Table 2) was applied to the collected data for the category workers.

Seven impact subcategories and 13 inventory indicators were assessed to determine the social performance of the workshops analyzed, which is an "output compared to a known standard, often expressed as a score" [36] (p. 25). The indicators were selected and adapted from the Methodological Sheets for Subcategories in Social Life Cycle Assessment (S-LCA) [44], taking into account the national legislation regulating labor relations and the cost of the basic food basket, so that they fit the informal context in which several of the selected materials are produced (Table 3).

Table 2. Pedigree matrix for social assessment data.

Data for Actor	Indicator	Results	Justification
Workers: producers of construction materials	Reliability of the source(s)	2	The data collected for this category of actors came directly from the producers selected to participate in the study, who live in the same area and, based on interviews, are the most experienced producers. These producers share the same steps in the production process, which is standardized and does not vary from workshop to workshop.
	Completeness conformance	3	The data are the result of the application of interviews and surveys with the most experienced producers in the area.
	Temporal conformance	1	The data were collected during the year 2022, with the intention of being able to evaluate the impacts throughout that year.
	Geographical conformance	1	The data correspond to the location determined in the objective and scope of the evaluation.
	Further technical conformance	1	The data correspond to the sector identified in the objective and scope of the evaluation.

Table 3. Impact subcategories and inventory indicators adapted to fit the informal context.

Impact Subcategories	Inventory Indicator
Fair Salary	Monthly income of the people working in the workshop
	Producers consider that their income is sufficient to cover their needs
Working Hours	Overtime hours worked per week
	More than 5 working days per week
Freedom of Association	Restriction on freedom of association
Child Labor	Children working in the workshop
Health and Safety	Use of adequate protective equipment
	Injuries related to the production process
	Negative impacts on health and safety due to harmful substances or the work environment
	Regular health check-ups for people working in the workshop
	Sanitary facilities in the workshops
Social Security	People working in the workshop with social security
Job Satisfaction	Satisfaction with the productive activity

In order to collect the necessary information for each of the selected indicators, questions were constructed for the questionnaire applied to the producers of the selected materials; for example, the impact sub-category of “fair salary”, which is composed of two inventory indicators, (a) monthly income of producers in the workshop and (b) producers consider that their income is sufficient to cover their needs.

For the first inventory indicator, (a) questions such as “How does your monthly income compare to the current legal minimum wage?” and “How much is your monthly income in dollars?” were included in the survey form. For the second inventory indicator (b), questions such as “Do you consider that your income is sufficient to cover your needs?”

were also included. The survey questions used to collect data for all indicators are detailed in Supplementary File S3.

The Type 1 (Reference Scale Approach) evaluation was based on the construction of reference scales. These were developed for each indicator used in the research (Table 4) and were also adapted to the context of the Ecuadorian construction sector, which is characterized by its informal nature. In this way, we ensured that the reference scales were adjusted to the concrete reality of the case study for which this research was carried out, without losing the connection with the set of general indicators that have been applied in a diversity of cases through the methodological guidelines of social life cycle analysis.

Table 4. Example of reference scales for the indicators used in the research.

Impact Subcategory	Inventory Indicator	Reference Scale
Fair salary	Monthly income of the people working in the workshop	2: Very high performance Income is above the average for producers with wages above USD 425 (unified minimum wage in Ecuador 2022).
		1: High performance Income is below the average of producers with a salary higher than USD 425.
		0: Neutral performance Wage is equal to that defined by law (USD 425 unified minimum wage in Ecuador 2022).
		−1: Low performance Income is above the average of producers with salaries lower than USD 425.
		−2: Very low performance Income is below the average of producers with wages below USD 425.
	Producers consider that their income is sufficient to cover their needs.	2: Very high performance The salary allows the producer to cover their needs with expenses greater than USD 724.39.
		1: High performance The salary allows the producer to cover his basic needs through the Basic Family Basket (The Basic Family Basket (CFB) is a set of goods and services that are essential to satisfy the basic needs of a typical household composed of 4 members, which for the year of the evaluation had a cost of USD 724.39 and included 75 products)
		0: Neutral performance The salary allows the producer to cover his basic needs through the “Canasta familiar vital en Ecuador 2022” (set of goods and services that allow a family of 4 members to be above the poverty line, this basket for the year of the evaluation had a cost of USD 514 and included 73 products).
		−1: Low performance The producer needs one extra salary to cover their basic needs through the “Canasta familiar vital en Ecuador 2022” (USD 514).
		−2: Very low performance The producer needs two extra wages to cover basic needs through the “Canasta familiar vital en Ecuador 2022” (USD 514).

The scales were based on criteria derived from specific regulations, best practices, and statistics, thus allowing to “value” positive and negative social performance. The reference

scales with linear scores were built based on the recommendations of the Guidelines for Social Life Cycle Assessment of Products and Organizations [34], and an adaptation was also generated to better correspond to the local context and the production sector. The range was set from -2 to $+2$, where (-2) represents the worst performance, zero indicates neutral performance (benchmark set based on compliance with regulations that apply to the evaluated activity), and the top level $(+2)$ corresponds to positive or ideal performance. It should be noted that, for specific indicators (overtime worked per week, more than 5 working days per week, child labor, and sanitary facilities in the workshop) the reference scale ranged from -2 to 0 , since the neutral point corresponded to the non-existence of a negative social condition.

For the performance assessment, which involved applying the reference scales to the collected inventory data, the data were consolidated into a unified database using Excel (spreadsheet software). This database allowed for an automated evaluation using reference scales (ranging from -2 to $+2$). The final table, containing inventory and performance indicators, served as the basis for creating heat maps, dashboards, and reports to present the assessment results.

2.3. Analysis of Cultural Aspects Through Ethnographic Methodology

To identify the cultural meanings and key characteristics of the types of production processes of the selected local materials, an ethnographic approach was applied to the producers of construction materials. Through this approach, the meaning of material things and of the activities [45] carried out by the producers during their everyday life was identified and analyzed.

The data used for the development of the ethnographic method were of a verbal and linguistic qualitative nature. These qualitative data were condensed in the discourses of the producers. Through these discourses, it was possible to access the meanings and specific cultural characteristics behind the production processes. This is because discourses express the symbolic and material conditions that give meaning [46] to the production activity and at the same time, they make possible the creation of the discourse itself.

These qualitative data were collected through open-ended-question interviews (Supplementary File S4), which allow better access to the concepts and notions that individuals use to make sense of their activities by constructing argumentative systems [45] or conceptual/theoretical schemes using these concepts and notions. In addition, interference of the researcher in the answers given by the interviewed actors is significantly reduced.

The construction of these systems followed the QUAGOL methodology for qualitative data analysis [47]. This process involves moving from literal interview transcriptions to narrative reports and then to conceptual schemes, which are tested by rereading the interviews to ensure their explanatory adequacy. Once the theoretical schema is validated, the coding phase begins, using the list of concepts from the schema as starting codes. This step further validates the concepts, by coding the interview transcriptions with the schema's concepts. Finally, a conceptual framework is built, structured around the narrative of the producers' discourse, connecting key concepts, to organize and give meaning to their productive activity.

2.4. Towards Qualitative Integration of the Sustainability Dimension

We proposed a connection between the three dimensions evaluated, this proposal was of a qualitative theoretical nature and was generated through a qualitative inductive methodological approach [48]. With this approach, patterns of relationships were analyzed and identified based on the results of the three methodologies. These patterns were then contrasted with the discourses analyzed through the ethnographic method [45], in order

to qualitatively deepen the relationships between the analyzed dimensions. We took this methodological approach as the integrated analysis of the dimensions of sustainability is at an exploratory level that still requires theoretical and methodological development.

Part of the development of this proposal for the connection between the three dimensions at an exploratory level implies, as a result, generating an identification of key elements in each of the dimensions. This would be a first step to develop further investigations that can quantify these relationships and identify correlations between the evaluated dimensions.

3. Results

3.1. Environmental Impact Assessment

The results of the application of the two methods for the determination of the environmental impact of the production processes of the six selected materials are presented below.

1. IPCC 2021—Global Warming Potential 100, measured in kg CO₂-eq/kg of material produced.

Figure 1 shows the potential climate change impacts associated with the production of both traditional and contemporary building materials, expressed in kg CO₂-eq/kg of material produced, calculated using the IPCC 2021 method. The production of these materials in Ecuador generates lower amounts of CO₂-eq compared to the data from other countries, as presented in Figures 2 and 3.

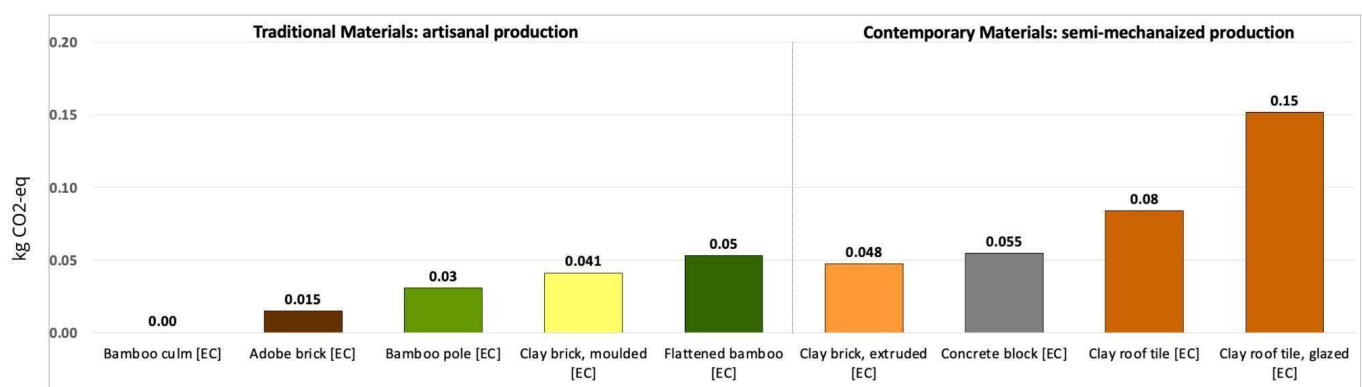


Figure 1. Potential environmental impact of 1 kg of material produced in Ecuador, by applying the IPCC 2021 method: EC = Ecuador.

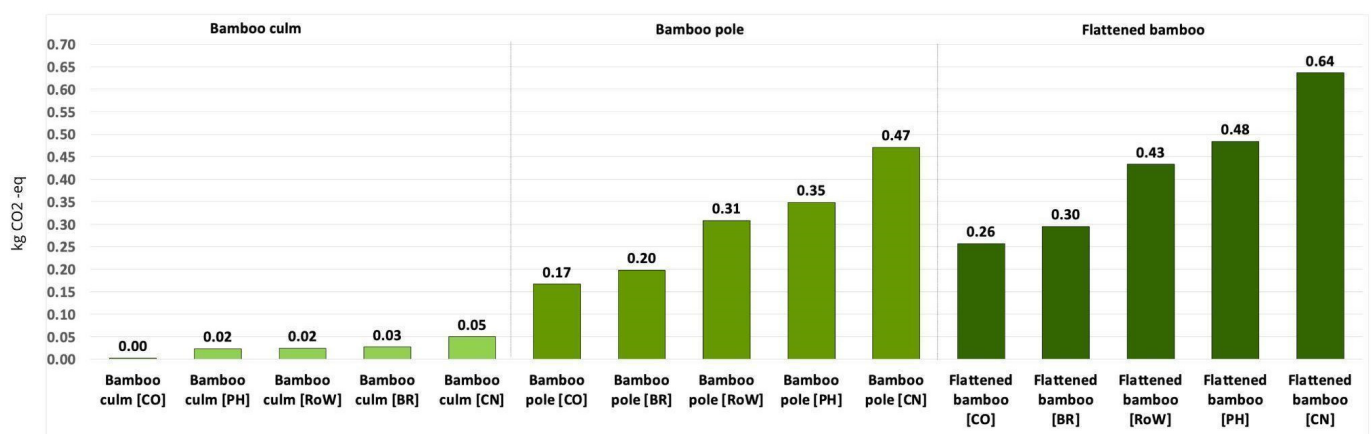


Figure 2. Potential environmental impact of 1 kg of bamboo pole, bamboo culm, and flattened bamboo produced in BR = Brazil, CN = China, CO = Colombia, PH = Philippines, RoW = Rest of the world, applying the IPCC 2021 method.

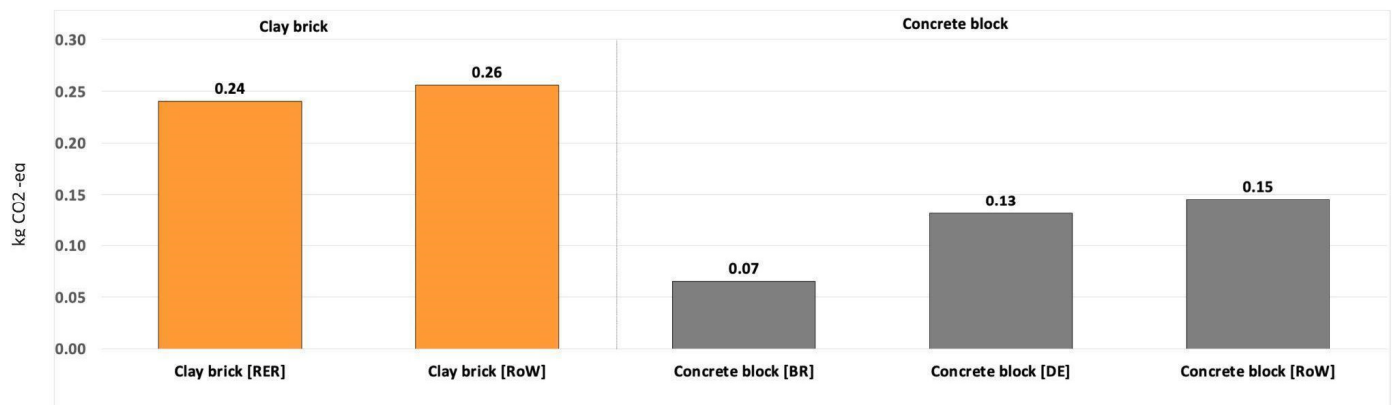


Figure 3. Potential environmental impact of 1 kg of clay brick and concrete block produced in RER: Region of Europe, BR = Brazil, DE = Germany, RoW = Rest of the world, applying the IPCC 2021 method.

In the case of bamboo culm, cane silviculture in Ecuador does not generate CO₂-eq emissions, since most of the crops are natural bamboo plantations. This is because the use of pesticides, herbicides, and fertilizers is uncommon, in addition to the fact that crop maintenance is carried out using artisanal and often manual techniques, avoiding the consumption of energy and the emissions from machinery. These factors contrast with production in China, where it is estimated that 0.05 kg CO₂-eq/kg of material produced is generated by the mechanization of the process and the use of fertilizers (Figure 2). The process of preserving bamboo poles in Ecuador shows significantly lower CO₂-eq emissions per kg of material produced compared to similar activities in other Latin American countries and World regions (Figure 2). Specifically, these emissions are 84% lower compared to Colombia and 94% lower compared to China. In addition, the global warming potential associated with flattened bamboo production in Ecuador shows lower scores compared to Colombia and China; in particular, it is 5.7 times lower than in Colombia and 13 times lower than in China. These results are attributed to the artisanal approach that prevails in the production of these materials in Ecuador, characterized by the minimal use of machinery, which contributes to a reduced carbon footprint. In addition, it is important to note that the current demand for these materials does not require considerable investment to implement more mechanized processes.

The global warming potential scores of adobe and molded clay brick production were compared with the those of brick production outside Latin American regions, since there are no LCI data on the production of this material available in ecoinvent for Latin American regions. The production of adobe and clay brick represents 6% and 16%, respectively, of the emissions of kg CO₂-eq/kg of material produced compared to the average European reference and the rest of the world (Figure 3). This is because the production of these materials uses firewood as fuel instead of fossil fuels, as is the case for the average European reference and the rest of the world.

Concerning the production of contemporary materials (Figure 1), the production of glazed tiles presents a higher amount of CO₂-eq emissions compared to the other materials; this is due to the additional firing after glazing, which leads to a higher use of fuel (firewood), thus corresponding to 53.3% of kg CO₂-eq emissions more than the production of unglazed tiles. The production of extruded clay brick in Ecuador presents 80% lower CO₂-eq emissions compared to the average European reference and the rest of the world (Figure 3).

In the production of concrete blocks, Ecuador has a lower CO₂-eq per kg of material produced compared to Brazil, Germany, and the rest of the world, with a difference of

29%, 62%, and 67%, respectively (Figure 3). The environmental impact of concrete blocks accounts for the embedded impact of the precursor processes, including the production of clinker and cement. Cement and transportation are the main flows contributing to climate change in Ecuador, with 39% and 44%, respectively. In Brazil, the largest contributing flow comes from cement, accounting for 75% (Figure 3). Therefore, transportation in Ecuador generates a greater amount of greenhouse gases and, consequently, a greater contribution to climate change compared to Brazil.

Generally, the low emission levels associated with the production of traditional materials are attributable not to the inherent properties of the materials themselves, but rather to the methods employed in their fabrication, specifically artisanal techniques. These methods eschew the use of machinery and, consequently, do not necessitate the intensive consumption of energy for operation. This distinction may also elucidate regional disparities; for instance, the “Rest of the World” (RoW) and the average European benchmarks exhibit outcomes indicative of more mechanized production processes.

2. Environmental footprint (EF), measured in mPt/kg of material produced.

Figures 4–6 show the weighted environmental impact of the production of traditional materials in Ecuador compared to other countries, calculated using the EF 3.0 environmental footprint method and expressed in mPt/kg of material produced. Each bar in the figure represents the weighted sum of the results for each impact category for each material.

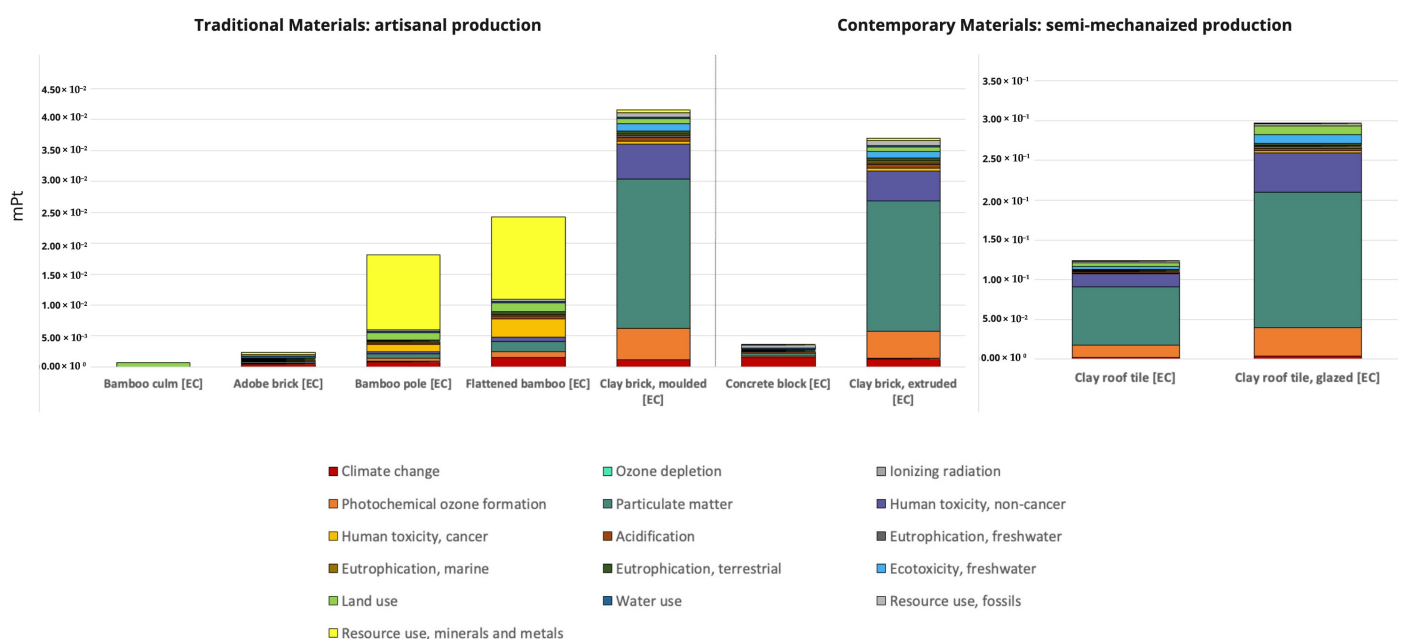


Figure 4. Weighted scores for the production of 1 kg of traditional and contemporary materials in Ecuador, according to the Environmental Footprint (EF) v.3.0 method. Note that the scores for clay roof tiles refer to a secondary scale, due to the significant difference in results. EC = Ecuador.

When comparing the potential environmental impacts of the production of traditional materials in Ecuador, both adobe and bamboo have a considerably lower environmental footprint compared to molded clay brick, which is the traditional material with the highest environmental impact. The environmental footprint of adobe is 94.5% lower than that of molded clay brick, while that of bamboo poles is 87% lower. The categories with the largest proportion in the environmental footprint of adobe production are climate change with 18% and water use with 19%. Climate change has this percentage because the weight assigned to this indicator was the highest of all (21%), and the rest of the indicators had a weight of less than 9%. In the case of water use, its high percentage was due to a combination of

the inventory factor and the weight assigned to the indicator (it was third in the list, with 8.5%).

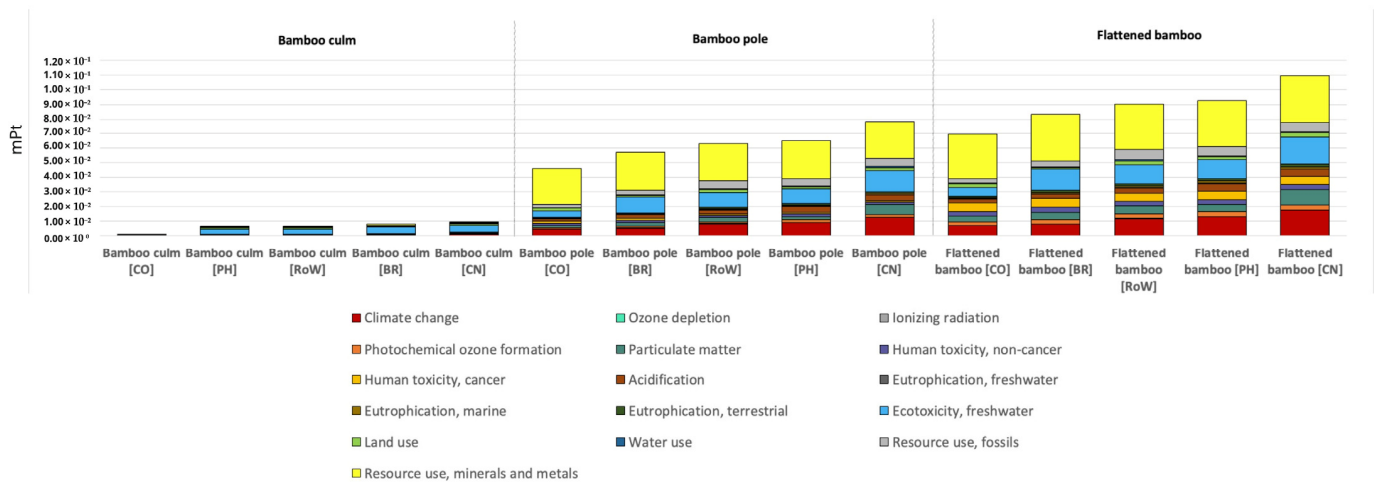


Figure 5. Weighted scores for the production of 1 kg of bamboo in different regions of the world, according to the Environmental Footprint (EF) v3.0 method. For the acronyms of the geographical regions, refer to Figure 2.

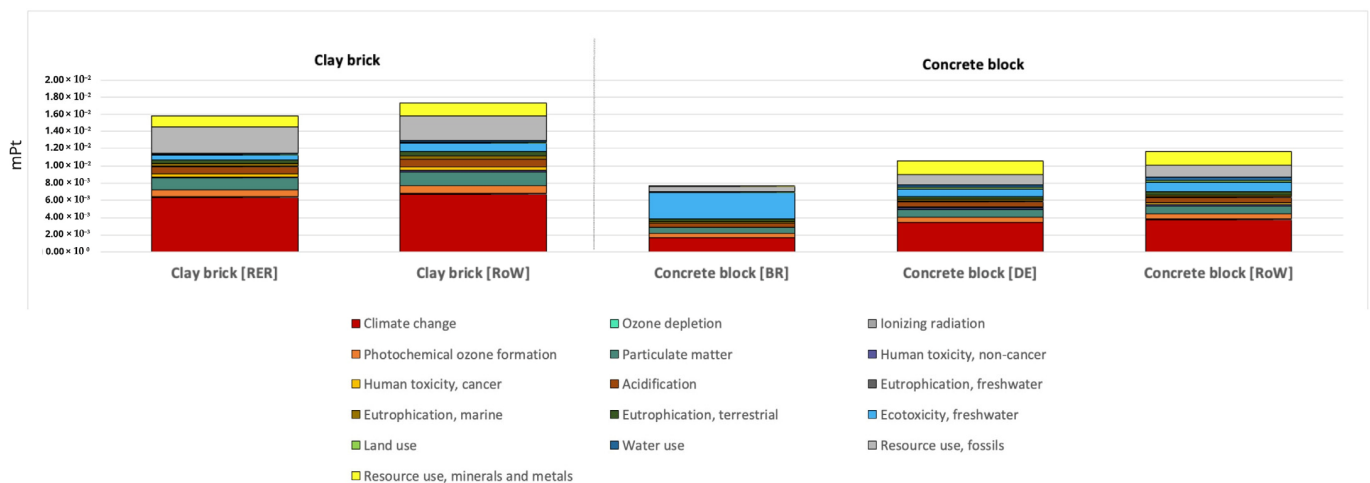


Figure 6. Weighted scores for the production of 1 kg of clay brick and concrete block in different regions of the world, according to the Environmental Footprint (EF) v3.0 method. For the acronyms of the geographical regions, refer to Figure 3.

Bamboo (culm) forestry in Ecuador (Figure 4) represents the activity with the lowest environmental impact, compared to the impact of the same material in other countries (Figure 5). The only category that generates environmental load in this activity is land use, with a value of 7.43×10^{-4} mPt/kg of material produced. This category represents 7% of the environmental footprint of the production of bamboo poles, of which bamboo culm is the main input. This impact is associated with land occupation due to the annual harvesting of bamboo culm. Other categories that represent a greater proportion of the environmental footprint of both bamboo poles and flattened bamboo are the use of resources, minerals and metals, and human toxicity (carcinogenic). These results are associated with the use of clay and the use of borax and boric acid in the process of preserving the canes, as well as the emission of ash from burning the canes to prepare the solutions. Comparing with other countries such as Colombia, Brazil, and China, their total environmental footprint of bamboo is significantly higher than that of Ecuador and its largest category is the use of mineral resources and metals (Figures 5 and 6). For bamboo poles, this category contributes

to 53% of the total environmental footprint in Colombia, 45% in Brazil, and 32% in China. For flattened bamboo, the figures stand at 43%, 38%, and 28%, respectively, for Colombia, Brazil, and China.

In the case of the production of molded clay brick in Ecuador (Figure 4), particulate matter, human toxicity (non-carcinogenic), and photochemical ozone formation are the categories with higher scores, representing 57%, 13%, and 12% of the total environmental footprint, respectively. These results are associated with the particulate matter emissions produced during the firing process, where wood combustion also generates carbon monoxide and other gases such as volatile organic compounds (VOCs) and nitrogen and sulfur oxides. Compared to the average European and global benchmark (Figure 6), the total environmental footprint of molded clay brick is significantly higher, approximately doubling its value. This is because the emissions of particulate matter represent a large proportion of the footprint, which are associated with the artisanal production process of this material, where there are no controls or mechanisms to reduce emissions and combustion is incomplete.

Regarding the environmental footprint of contemporary materials, according to Figure 4, glazed tile has the largest environmental footprint with 2.97×10^{-1} mPt/kg of material produced, followed by tile with 1.23×10^{-1} mPt/kg of material produced and extruded clay brick with 4.78×10^{-2} mPt/kg of material produced, due to the consumption of diesel.

In the production of glazed tile, particulate matter has a significantly large environmental load compared to other construction materials, and this is associated with the varnishing process, due to the use of lead oxide and the burning process, reaching 1.71×10^{-1} mPt/kg of material produced, which for regular tile (without varnishing) is 7.31×10^{-2} mPt/kg of material produced.

In concrete block production in Ecuador (Figure 4), the categories with the largest proportion of the environmental footprint are climate change, use of mineral resources and metals, and particulate matter, representing 42%, 14%, and 10% of the total footprint, respectively. These results are attributed to embedded pollution in the block's precursor processes, i.e., clinker and cement production. During these processes, heavy metals, sulfur oxides, nitrogen oxides, VOCs, and particulate matter are emitted, contributing to the carbon footprint and the presence of particulate matter in the air. In addition, the raw materials used, such as limestone, clay, gypsum, sand, and gravel, are directly related to the mineral resources and metals use category. Compared to other countries (Figure 6), Ecuador's environmental footprint is smaller compared to Brazil, Germany, and the rest of the world, corresponding to 53%, 65%, and 69%, respectively. Climate change in Ecuador has an environmental burden approximately twice as high as in Brazil, while freshwater ecotoxicity shows a much higher impact compared to Ecuador, where it is 98% lower.

In the production of extruded clay brick, the categories with the largest proportion of the environmental footprint are particulate matter and photochemical ozone formation, due to the emission of sulfur and nitrogen oxides during the combustion of firewood in the brick firing process. These emissions increase the formation of tropospheric ozone (smog) and increase the ash load (particulate matter) in the environment. As for the comparison of the environmental footprint between the two types of brick production in Ecuador, brick where only diesel consumption is involved is 52% higher than the hybrid process (diesel and electricity consumption); it also had a higher load in all the impact categories evaluated. Compared to the average European reference (RER) and the rest of the world (RoW) (Figure 6), the extruded clay brick from Ecuador shows a significant difference, being 53% higher in terms of total environmental footprint. In particular, the emission of particulate matter in Ecuador is approximately 14 times higher than these international references, while the carbon footprint is 5 times lower.

3.2. Social Impact Assessment of the Production of Construction Materials

This section presents the results of the social impact assessment related to the production of local building materials, which were divided into two categories: traditional materials produced through artisanal methods (adobe, molded clay brick, and bamboo) and contemporary materials produced using semi-mechanized processes (extruded brick, tile, and concrete block).

Some common characteristics were identified in the analyzed workshops. In the adobe and molded clay brick workshops, it was found that it is common for a single person to carry out almost the entire production process, with the help of members of the family circle with specific processes of material transfer. The same person also owns the workshop and the tools necessary to produce the materials. In the case of bamboo, it was found that much of the *guaduales* (bamboo plantations) from which the cane is extracted is of natural origin and was inherited by the current owners. This means that the plantation's growing phase has no planned process or investment. It was also found that an average of three people is needed to extract bamboo, which can vary depending on the size of the plantation. In the workshops of the contemporary semi-mechanized production materials, a significant shift in labor dynamics was observed. Indeed, the support of the family circle was drastically reduced, and the number of hired people increased. On average, five people work in the semi-mechanized workshops, which was concentrated mainly on transferring materials within the workshop and not so much on the manual transformation of the raw material.

Unlike the environmental life cycle impact assessment, for which the results were presented by material, the social life cycle impact assessment results are presented in tables of social performance for each material production workshop (Table 5).

Table 5. Social performance for traditional materials and contemporary materials.

Impact Sub-categories	Inventory Indicator	Traditional Materials Artisanal Production				Contemporary Materials Semi-Mechanized Production		
		Adobe	Molded Brick	Bamboo		Extruded Brick	Clay Roof Tile	Concrete Block
				Guadual	Preservation Center			
Fair Salary	Monthly income of the people working in the workshop.	−2	−2	0	0	0	0	0
	Producers consider that their income is sufficient to cover their needs.	−2	−2	2	2	0	0	2
Working Hours	Overtime hours worked per week	−2	0	0	0	0	0	0
	More than 5 working days per week	−2	−1	0	0	0	1	0
Freedom of Association	Restriction on freedom of association	0	0	0	0	0	0	0
Child Labor	Children working in the workshop	0	0	0	0	0	0	0

Table 5. Cont.

Impact Sub-categories	Inventory Indicator	Traditional Materials Artisanal Production				Contemporary Materials Semi-Mechanized Production		
		Adobe	Molded Brick	Bamboo		Extruded Brick	Clay Roof Tile	Concrete Block
				Guadual	Preservation Center			
Health and Safety	Use of adequate protective equipment.	−2	−2	2	2	2	2	2
	Injuries related to the production process.	−2	−2	0	0	1	0	0
	Negative impacts on health and safety due to harmful substances or the work environment	1	0	0	0	0	0	0
	Regular health check-ups for people working in the workshop	−2	−2	−2	−2	−2	−2	−2
	Sanitary facilities in the workshops	−2	2	2	2	2	2	2
Social Security	People working in the workshop with social security	−2	−2	−2	−2	2	2	−2
Job Satisfaction	Satisfaction with the productive activity	2	2	2	2	2	2	2
−2	−1	0			1		2	
Very low performance	Low performance	Neutral performance			High performance		Very high performance	

3.2.1. Social Performance of Traditional Materials Produced Through Artisanal Methods

In the case of adobe production workshops, we can observe that most indicators had the worst negative performance (−2), specifically in the subcategories of fair salary, working hours, health and safety, and social security. This is because the monthly income of adobe producers is around USD 90 per month. As for working hours, the workers and producers spend on average 10 working hours per day in the workshop and generally have a 7-day work week. For the indicator “Negative impacts on health and safety due to harmful substances/working environment” in the health and safety impact subcategories, a positive performance was recorded; this is because no chemical or toxic substances of any kind are used to produce adobe.

The inventory indicator “Satisfaction with the productive activity” from the subcategory “Job Satisfaction” expressed an interesting finding, which surprisingly showed the highest positive level (2). At first, this may seem contradictory, but upon delving into the socio-cultural meanings, a profound cultural motivation was discovered that keeps producers engaged in this type of production despite the negative performance in other

impact subcategories. For these producers, adobe production is not just seen as the creation of construction material but also as a means of preserving and passing knowledge from their ancestors and community.

In the case of molded clay brick production workshops, the worst negative performance (−2) was shown for the fair salary indicators, with an average monthly wage of USD 220 across the workshops analyzed. Regarding working hours, the performance improved slightly, ranging from −1 to 0, due to the reduced human labor required for molded brick production; unlike adobe, this material has a burning phase. This reduces the drying time and, therefore, the time required by the producer for this phase. At the same time, the burning phase generates emissions that can be harmful to health, affecting the health and safety impact subcategory, especially for the indicator “Negative impacts on health and safety due to harmful substances or the work environment”, which no longer had the same positive performance as in the case of adobe. The rest of the indicators remained the same as the performance of adobe, since the production process of molded brick does not vary significantly, except for the inclusion of the burning phase mentioned above. Therefore, the production of molded bricks in the studied area has a similar negative impact on the quality of life of the workshop producers. Regarding the subcategory of job satisfaction, we can find the same feature mentioned for adobe. The highest positive level (2) of performance, despite the negative performance in the other impact subcategories.

To understand the results presented in the performance table corresponding to bamboo, it is essential to mention a significant difference between the processes required to produce a bamboo pole, adobe, or a molded brick (Supplementary File S1). This difference primarily lies in the amount of human labor required to produce each material. In the case of bamboo, we are mainly talking about the process of extraction and preservation of the raw material, in which no significant transformation of the raw material is performed. In the case of adobe and molded brick, the opposite is true, since the raw material undergoes an intensive transformation process in order to be considered a finished building material.

This difference is clearly expressed in the column corresponding to the gradual (bamboo plantation), where the extraction process is carried out. The amount of human labor required in this process is lower than the amount needed to produce adobes or molded bricks. In the case of the gradual analyzed, it was even more reduced, since it is not a planned plantation, but left to grow naturally. Therefore, the performance in the subcategory of working hours was neutral. For the gradual analyzed, a maximum of eight hours per day and three days a week are necessary to complete the extraction process of 24 bamboo poles (what the owners of the gradual call a “balsa”); this can vary according to demand.

In the case of the fair salary subcategory, the performance was neutral, equal to the minimum wage established by law in Ecuador (USD 425) for 2022. In the Canton of Portoviejo, Manabí Province, the International Bamboo and Rattan Organization (INBAR) has played a significant role. They have provided training and technical advice to owners of natural bamboo plantations and conservation centers, leading to an improvement in the quality of bamboo canes and an increase in their selling price. This has resulted in a higher income for the owners and workers of the bamboo plantations and conservation centers. In addition, INBAR has also trained construction workers, to create technical capacity in the area to use bamboo as a construction material for housing. This creates greater demand, which also favored the performance closer to neutral in the fair salary subcategory.

3.2.2. Social Performance of Contemporary Materials Produced Through Semi-Mechanized Processes

As illustrated in Table 5, the performance across the three workshops dedicated to contemporary material production was notably similar. This uniformity is largely attributable

to the integration of machinery in the production process, which reduces the reliance on manual labor, while simultaneously enhancing the production capacity. An additional advantage of these materials, not immediately evident from the data presented in the table, lies in their high regional demand. This factor contributes to increased profitability, which, in turn, supports improved outcomes regarding fair wages. Moreover, this heightened demand can lead to an extension of the workweek, surpassing the standard five working days mandated by law when production capacity is exceeded. This general analysis underscores the significant impact of mechanization on workshop performance.

In the case of extruded clay brick production workshops, Table 5 shows values that express a neutral performance in the subcategories of fair salary, working hours, and freedom of association. The performances for fair salary and working hours are an effect of the mechanization of the production processes, particularly the mechanization of the mixing and molding stage. As mentioned above, mechanization reduces the workload, increases production rates, and allows for greater economic profitability, which, in combination with the high demand for this material, allows greater profitability for the workshop, which is expressed in an increase in their social performance.

For the clay roof tile production workshops, in general, the values in the social performance table express a neutral performance in the sub-categories of fair wages, working hours, freedom of association, and child labor. In the specific case of this material, the molding stage of the production process is also semi-mechanized.

A similar scenario for the remaining contemporary semi-mechanized production material can be identified. When analyzing the concrete block results, the values mostly express neutral performance in the subcategories of fair wages, working hours, freedom of association, and child labor. In the specific case of the production of this material, the mixing and molding stages have been mechanized.

3.3. Cultural Meanings in the Production of Construction Materials: A First Step Towards the Cultural Dimension of Sustainability

In this section, the results obtained through the application of the ethnographic method are presented, revealing key cultural meanings of the production of building materials linked to the circulation of techniques and knowledge essential for artisanal production, which also made it possible to start semi-mechanized production of some of the traditional materials. These techniques and knowledge are passed down through generations within communities and families, thus being an essential part of their identity. It is important to mention that the results presented in this section do not follow the same logic of impacts as in the case of the E-LCA and the S-LCA. The meanings identified through the ethnographic method are rather elements that could be impacted by directly affecting the identity that producers have created around the activity of construction material production.

Figure 7 presents a theoretical schema of the material production process, constructed from common elements in the producers' discourses. This scheme allows us to organize the concepts and meanings that the producers use to make sense of artisanal and semi-mechanized production.

This diagram shows the relationships between the three main components that make up a building material production process, whether it is artisanal or semi-mechanized. These three components are as follows:

- Habitat needs: The inhabitants of a territory have a need to create habitable spaces.
- Resources available: The natural resources available in the area.
- Techniques, knowledge, and skills: They allow the transformation and use of the available resources to create habitable spaces.

The relationship between these three components is developed as follows. The need to create habitable spaces meets the available natural resources and generates techniques, knowledge, and skills to transform and use these resources. This allows the generation of production processes building materials to satisfy the initial need to create habitable spaces. In this way, the resources available strongly influence the type of techniques, knowledge, and skills that are developed and, therefore, the type of building materials that can be generated.

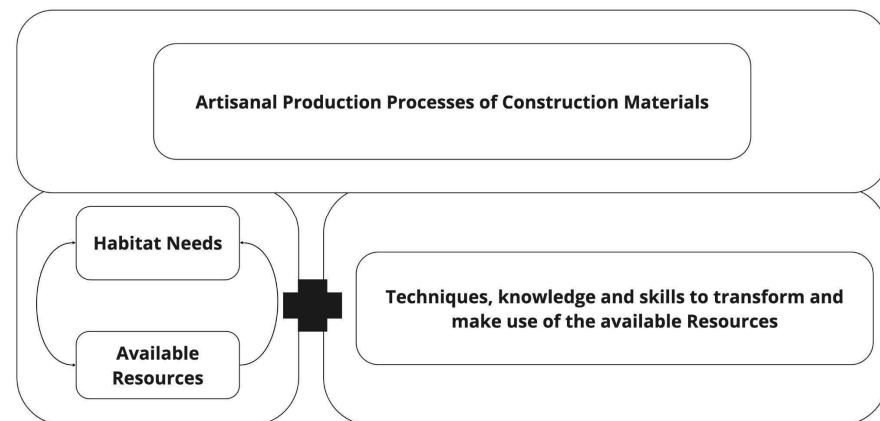


Figure 7. Theoretical scheme of the discourse of the producers.

These three components combine tangible and intangible elements that are key to an in-depth understanding of building material production processes in general. The content of each component will vary as producers use different concepts to give cultural meanings to their activity according to the type of production, artisanal or semi-mechanized production. The section below presents the meanings identified within each component for each type of production, following the logic of Figure 7.

3.3.1. Artisanal Production: Traditional Materials (Adobe, Molded Clay Brick, and Bamboo)

The scheme is presented below with the information of the artisanal production of traditional construction materials (Figure 8), along with a description of the meanings that comprise it.

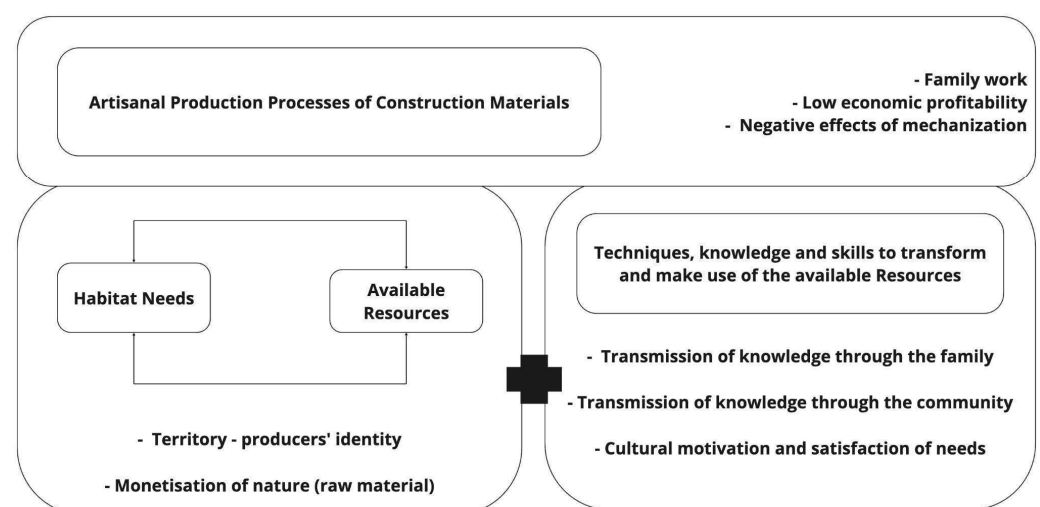


Figure 8. Unified scheme of the concepts that give meaning to the artisanal production process.

We will start with the two lower components of the scheme, as these two form the basis for the consolidation of the material production process.

In analyzing the relationship between habitat needs and available resources, two key concepts emerged from the discourses of artisanal producers: territory-producers' identity and the monetization of nature (raw materials). Hector, a semi-mechanized tile producer, highlighted that Racar, a sector of the Sinincay parish in the city of Cuenca, has been a site for producing construction materials for over 100 years. Additionally, his possession of an old oven, which was originally owned by his wife's great-grandparents—who were also engaged in this productive activity—further supports his perception of Racar as an industrial zone for the production of construction materials. Luis, an artisan brick producer, noted that all producers use the same materials and live near their workshops, showing a strong connection with the territory, where generations of their families have also been engaged in construction material production.

The territory-producers' identity: This implies that producers have a bond with the area in which they work, since it is a space that has taken shape as a function of the activity they carry out. Likewise, the materials they produce take shape based on the resources available in that territory.

The monetization of nature (raw materials): This refers to how producers access raw materials and how this has changed over time. Currently, producers acquire materials through traders at prices set by intermediaries, leading to a disconnection with the extraction process. Previous generations, including parents and grandparents, had direct access to local raw materials, without monetary exchange, which facilitated the growth of artisanal production, as access to these materials did not involve costs for the producers. Now, raw materials often come from external sources, raising production costs and threatening the survival of artisanal practices like adobe and brick production.

The need to create habitable spaces is closely tied to available resources, fostering the development of techniques, knowledge, and skills to transform and use these resources. These techniques are formalized and passed down through generations of producers. The knowledge possessed by producers regarding the utilization of these resources is present in their discourses, allowing for the identification of the concepts presented below.

Transmission of knowledge through the family: Techniques and skills for producing building materials have been transmitted/taught between family members, from one generation to the next, through direct practice in the workshops.

Transmission of knowledge through the community: Knowledge is not only limited to family circles, it also circulates in the community where these production processes take place. In the case of this study, the knowledge to produce adobe and molded brick is common knowledge in Racar and San Jose de Balzay, both sectors of the Sinincay parish in the city of Cuenca, and, as we will see in the part on semi-mechanized production, it has served as a basis for the development of semi-mechanized production.

Cultural motivation and the satisfaction of needs: Despite limited income from artisanal production, producers continue this work because it allows them to apply knowledge passed down through generations. Their identity as artisanal producers, strongly tied to the territory, is reinforced through this production process, serving as a key motivation. Their initial motivation does not stem from a cost-benefit or investment logic but from the need to create habitable spaces for the community. Once this need is met, the techniques for transforming the available resources are used to generate economic income for the producers. This relates to the last component, the production process of artisanal materials itself.

The final component focuses on the artisanal production process of the three traditional materials analyzed. At this level, the production of artisanal building materials is

consolidated and no longer solely depends on the need to create habitable spaces for the producers themselves. It is also driven by the need to generate economic income from the knowledge they have developed about the resources available in their territory. The key concepts identified in the producers' discourses that characterize artisanal production are outlined below.

Family work: Artisanal production is largely based on family work. Most of the people involved in the production process are relatives of the workshop owner. This is directly related to the transmission of knowledge through familial links and the community.

Low economic profitability: The social performance tables for adobe and molded clay brick show that producers do not generate enough income to meet all their needs. Interviews revealed that this low profitability is mainly due to a lack of demand and competition from contemporary materials. As a result, some producers engage in other activities, such as subsistence farming, to meet their needs. This also limits their ability to hire workers, increasing dependence on family labor and affecting the continuity of this type of production.

Effects of mechanization: This concept is present in the two types of production, which generate different impacts. In the case of artisanal production, the effect is negative, since it generates unequal competition in terms of production rates, workload, and prices. An example of this negative effect is the mechanization process of artisanal brick and tile production (molded clay brick and semi-mechanized tile), which has generated a displacement of artisanal production forms.

In the case of adobe production, there is only non-mechanized artisanal production. Therefore, the material does not compete with other forms of production. In this way, it still has a significant cultural value, as it is a totally handmade work.

3.3.2. Contemporary Materials: Semi-Mechanized Production (Extruded Clay Brick, Clay Roof Tile, and Concrete Block)

Following the same logic of analysis, the main scheme is presented below with the information of the semi-mechanized production of contemporary materials (Figure 9), along with a description of the meanings that comprise it.

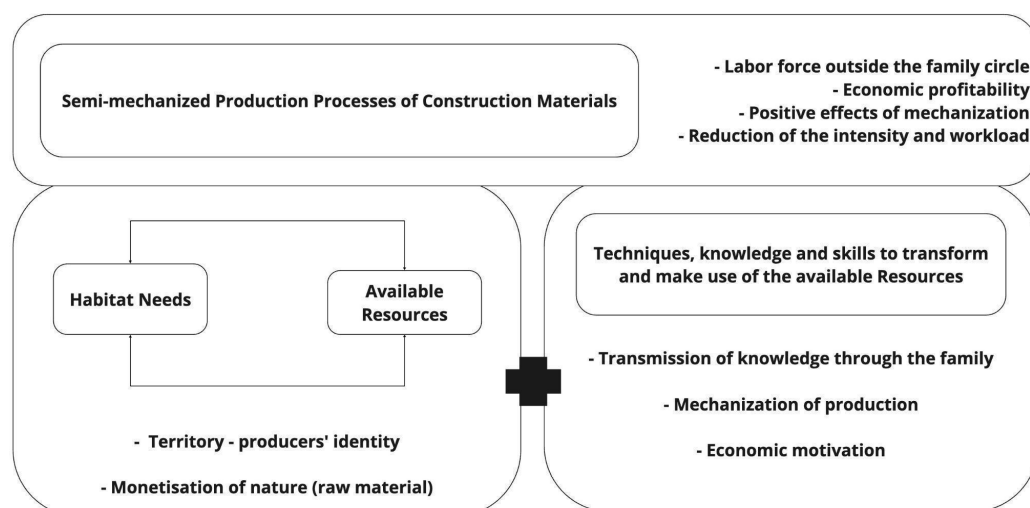


Figure 9. Unified scheme of the concepts that give meaning to the semi-mechanized production process.

In semi-mechanized production of contemporary materials, the relationship between habitat needs and available resources incorporates the same key concepts as in artisanal production, but with distinct definitions, which are outlined below.

Territory and producers' identity: In this research, we identified this concept among producers who use semi-mechanized brick and tile production. Unlike artisanal production, their identity is no longer directly tied to the natural resources of the territory but is instead mediated by artisanal knowledge passed down from the previous generation. This knowledge, inherited from their parents, is highly valued by the owners of semi-mechanized production workshops.

In the semi-mechanized tile production workshops, we identified that the producers not only use the knowledge of artisan production to achieve the correct mixture of the different types of soil. They also maintain the traditional design of the tile. Based on the knowledge of artisan production, the producers of this material have incorporated machines into their production processes, generating a continuity in relation to the application of artisan knowledge for production.

This is the last connection between the two types of production and the only mechanism that generates identity within the territory for producers of contemporary materials. If we compare the production of roof tiles with the production process of concrete blocks, we see that the semi-mechanized production process of concrete blocks demands a type of machinery, materials, and knowledge that increasingly distances the producer from his immediate territory, and from the techniques and knowledge that circulate within it, thus weakening the generation of an identity that is only possible thanks to the circulation of knowledge that arises from a relationship between artisanal producers and available resources.

Monetization of nature (raw materials): In the case of this type of production, the relationship with nature (raw materials) is mediated entirely by monetary transactions. This directly distances the producers from the available natural resources, making it difficult, once again, to generate an identity under the same logic that artisanal production allows.

For the bottom right component, the concepts associated with this type of production are directly related to the integration of machinery into the material production process and are outlined below.

Transmission of knowledge through the family: In analyzing the interviews of both artisanal producers and those with semi-mechanized production, we identified a common element: the circulation of knowledge related to artisanal production in San José de Balzay, part of the Sinincay parish in Cuenca. This knowledge transmission primarily occurs within families. All interviewed producers operating semi-mechanized workshops in San José de Balzay come from a lineage that utilized artisanal techniques for producing construction materials.

This connection is evident in the progression from adobe to molded clay brick and finally to extruded clay brick, highlighting that the foundational knowledge of adobe production is crucial for developing these materials. The concept of familial transmission of artisanal knowledge indicates that semi-mechanized production does not entirely break from its artisanal roots. There is, for the moment, a very weak connection between the two types of production and, more importantly, a generational connection between the two types of producers. The weakness of this connection lies precisely in its generational aspect, since when the generation of artisanal producers disappears, this knowledge will cease to circulate. This would give way to the circulation of the knowledge necessary for semi-mechanized production.

Once the concepts that semi-mechanized production shares with artisanal production have been described, the concepts specific to this type of production are presented below. These are key to understanding how producers give meaning to their productive activity and how they justify the incorporation of machines into the production process

Economic motivation: In mechanized production, motivation is primarily driven by economic profitability, contrasting with the cultural motivations found in artisanal production. The efficiency gained from incorporating machines justifies the producers' continued investment in this production method. Several interviewees noted that the introduction of machinery in brick and tile production has enabled them to sustain their activities; without it, they would have had to seek alternative sources of income.

Mechanization of production: This second concept, specific to semi-mechanized production, has a strong link with economic motivation, since it allows solving some of the problems that arise in artisanal production and which do not allow it to be sufficiently profitable. On this basis, the mechanization of production not only has a direct effect on the production process and its rhythms, it also changes producers' meanings and motivations to carry out and maintain this productive activity.

For the final top component, the following concepts will help us to understand and characterize the semi-mechanized production process as a whole.

Effects of mechanization: The introduction of machinery for producing clay bricks, clay roof tiles, and concrete blocks has three primary effects. First, it displaces traditional fully manual techniques. Second, it reduces the amount of human labor needed for production. Third, it accelerates production speed, increasing the workshop's overall capacity. These effects are essential for understanding the various concepts identified in the discourses of producers who operate semi-mechanized workshops, as they directly arise from the use of machines.

Intensity and workload: In this type of production, the amount of human labor required to produce the materials analyzed is reduced thanks to the mechanization of the molding stage. This makes it possible to produce more elements in less time and with less labor.

Economic profitability: The reduction in labor and the increase in production capacity due to machinery significantly enhance the economic profitability of workshops compared to artisanal production. For instance, an adobe producer earns around USD 90 per month, primarily limited by the three-month production time and low demand. In contrast, concrete block producers earn approximately USD 425 monthly, benefiting from a much faster production rate, as the mixing and molding processes are mechanized, allowing for completion in just four days. Additionally, this increased productivity aligns with the demand for contemporary building materials, enabling workshop owners to recover their investments more quickly. Thus, the advantages of mechanized production lie not only in reduced production times, but also in the rapid market circulation driven by high demand.

Labor force outside the family circle: Increasing economic profitability allows hiring labor outside the family circle. In contrast to artisanal production, which relies heavily on family cooperation, semi-mechanized production drastically reduces the presence of the family. As a result, the workshop as a space for transmitting knowledge of artisanal production through the family is severely weakened.

4. Discussion

The objective of this study is to provide insight into the sustainability of traditional and contemporary construction material production in Ecuador. To this end, a comprehensive environmental, social, and cultural analysis of the production of these materials was conducted through an interdisciplinary lens. This approach not only presented the results of each methodology (Section 3), in this section, we also discuss the relevance of the results and identify potential interactions between the dimensions of sustainability under analysis (environmental, social, and cultural).

The results obtained from the environmental analysis (Section 3.1) clearly show that, when we compare the semi-mechanized production of contemporary materials with the artisanal production of traditional materials, also considered vernacular materials, these materials generate a lower environmental impact. This is in line with what has been stated in other studies, where vernacular materials, which are locally available, with lower transport requirements and lower embodied energy due to simpler production processes, can contribute to reducing environmental impacts [11–15].

In addition, the generation of environmental inventories for the six local materials analyzed allowed avoiding the under- or overestimation of impacts when using the production data of a material in one country to assess the environmental impacts of the production of the same material occurring in another country [18]. This increased the datasets available for Ecuador and Latin America, as well as the possibilities of generating environmental impact analyses according to the reality of the region.

On the other hand, the results of the social analysis (Section 3.2) add to several investigations that have applied the S-LCA methodology to the production of building materials [20,21,49] (although with limited materials, e.g., concrete, steel, and bamboo), presenting the social performance of four materials that had not been evaluated before in the region (adobe, molded clay brick, extruded clay brick, and tiles). The study demonstrates that S-LCA enables a comprehensive integration of social dimensions associated with the well-being of diverse stakeholders into a more holistic sustainability analysis. This approach extends beyond the conventional focus on environmental, economic, and aesthetic factors, which have traditionally shaped the selection of building materials [50].

Finally, the cultural results (Section 3.3) allowed us to identify meanings that follow the approach that architecture and the elements that make it technically possible, including the production processes of building materials, cannot be reduced to a pure functional form [23]. Behind each material production process and the houses built with these materials, there is a system of cultural meanings that give sense to the form (materials and houses), i.e., the material production processes and architecture as such represent a middle ground between form and culture [23].

The interaction of the results can be expressed in the form of trade-offs or synergies [24], with the objective of avoiding burden shifting, meaning moving impacts from one dimension to another, or enhancing impacts in several dimensions at the same time.

By integrating the findings of the environmental LCA (carbon footprint), the social LCA (social performance), and the identified cultural meanings, we present a preliminary relation between environmental impacts, social impacts, and cultural meanings. To illustrate this, the results of the molded clay brick (artisanal production) and the extruded clay brick (semi-mechanized production) are used as case studies following an inductive approach.

The results of the two types of life cycle assessments, categorized by production method, reveal that while carbon emissions are lower in molded clay brick artisanal production (Figure 1) and the impact on human well-being is notably higher (Table 5). Conversely, in extruded clay brick semi-mechanized production, there is an increase in emissions (Figure 1), but a decrease in the impact on human well-being, resulting in a more performance-neutral outcome (Table 5). Additionally, mechanization reduces the necessity for local knowledge, thereby undermining the continuity of knowledge transmission and the cultural identity tied to artisanal production.

This integrated assessment indicates a clear trade-off [24] between social and environmental dimensions, where the direction of this trade-off is determined by the type of production. The positive quality of artisanal production (reduction in emissions) is mainly possible by enhancing its negative quality (high impact on producers' well-being). The

positive quality of semi-mechanized production (reduced impact on producers' well-being) is mainly possible by enhancing its negative quality (increased emissions).

To gain a deeper understanding of the cultural aspects and to understand how these combine with the trade-offs mentioned above, two key cultural meanings within the artisanal production process were considered. These were the transmission of knowledge through the family and the concept of family work. These two concepts allowed the identification of a potential cultural impact, which can be observed in the effect that the production of concrete blocks has on the conditions that enable the production of artisanal materials and the identity of the producers themselves.

The semi-mechanized production process of concrete blocks requires specific machinery, materials, workers, and knowledge that increasingly separate the producer from their immediate territory and the techniques and knowledge that circulate within it. This consequently weakens the generation of an identity that links producers with the territory and the available resources.

This raises the question of whether the semi-mechanized production process is having an adverse effect on the cultural identity of the artisanal producers. The presence of semi-mechanized production has a considerable effect on the viability of traditional methods of material production, particularly those that do not utilize earth, such as the production of concrete blocks. But the focus should not be solely on the cultural impact that the mechanization of production processes can generate. It is also important to take into account the benefits that this may have, especially related to reductions in workload and reductions in ergonomic risks.

In addition to the evident impact of semi-mechanization on artisanal production, several spaces for the transmission of knowledge related to this type of production are also affected. As previously discussed for the concept of the labor force outside the family circle, semi-mechanized production significantly diminishes the role of family work. Consequently, the workshop, which traditionally served as a conduit for transmitting knowledge about artisanal production through the family, is now severely compromised.

In the case of semi-mechanized production materials using earth, there is a lower impact, since there is still a connection through knowledge that allows transforming the raw material into a construction material. As mentioned before, several producers of tiles and extruded brick (semi-mechanized production) use knowledge of artisanal production to achieve the right mix of soil types to produce the materials.

However, it is yet unclear how these prospective cultural impacts may interact with the social and environmental dimensions. A synergy between the environmental and cultural dimensions (Figure 10) was identified in the analyzed workshops. A synergy that occurs due to the use of artisanal production of construction materials in which the circulation of knowledge generated in the territory is maintained. This gives identity to the producers and allows the transformation of available resources into construction materials, generating a lower amount of emissions than those generated in semi-mechanized production processes. This should be considered, including the trade-off between the environmental and social dimensions mentioned in the previous paragraphs, since the performance of artisanal workshops is far below a neutral performance (Table 5).

When this relation between the environmental and cultural dimensions occurs through semi-mechanized production, a different effect is generated. The trade-off between the environmental and social dimensions mentioned at the beginning of this section should also include the possible impacts on the cultural dimension (Figure 11). Since the usefulness and circulation of the knowledge generated in the territory is drastically reduced, which directly affects the conditions that make possible the generation of the identity of the artisanal producers. This is due to the use of machines that are a key part of the production of

materials such as concrete blocks, machines that demand a knowledge that is not generated in the territory and does not correspond with the identity of the artisanal producers.

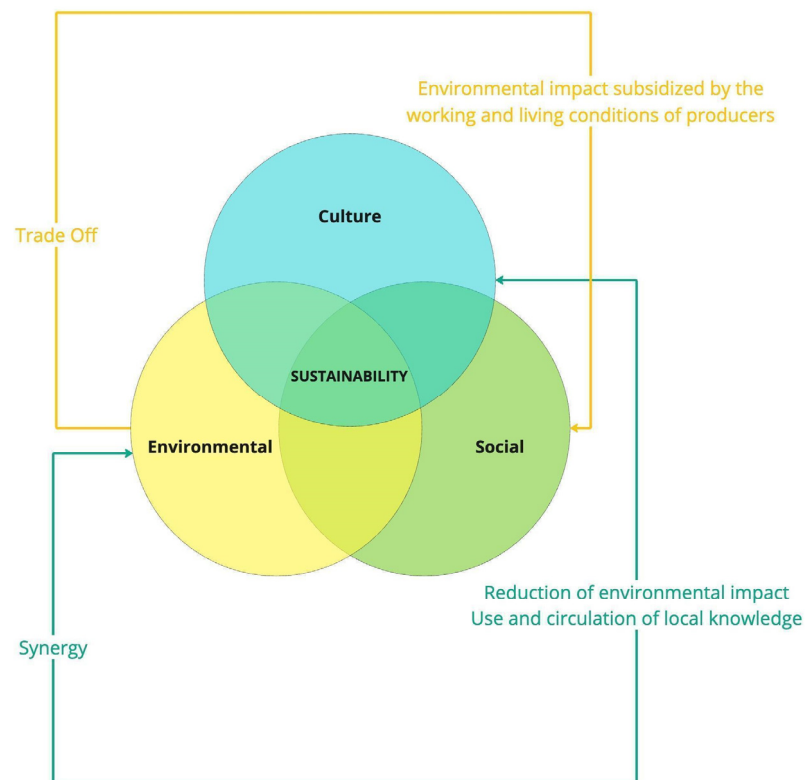


Figure 10. Trade-off and synergies between dimensions in the artisanal production of traditional construction materials.

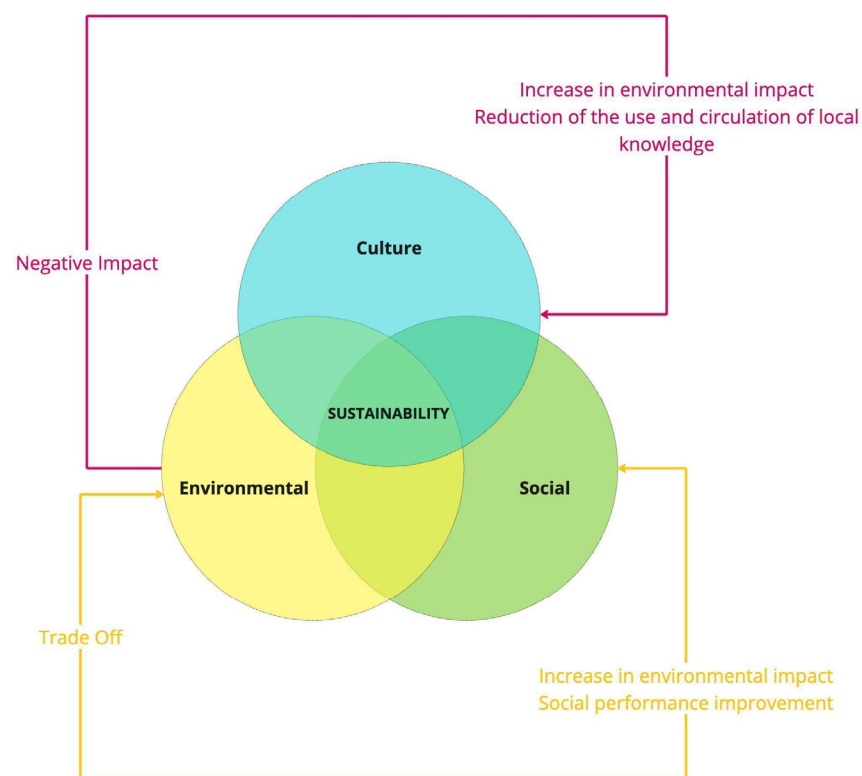


Figure 11. Trade-offs and impacts between dimensions in the semi-mechanized production of contemporary construction materials.

The results obtained and the connections identified between the different dimensions of sustainability were only possible thanks to two key elements of the project. First the collection of local data, which was vital in the construction of environmental and social life cycle inventories and the producers' discourses for the ethnographic analysis. This local data provided a profound understanding of the sustainability of these processes at the local level, with significant territorial relevance. The second key element was the use of an interdisciplinary approach, which allowed us to put different types of data into dialogue to explain the same phenomenon. This also implies a complex interpretation process, since we do not talk about the social, environmental, and cultural dimensions in isolation. We speak simultaneously about all three and how they interact with each other at a local level.

It is precisely the combination of all these elements that is the approach that this research proposes to address the complexity of sustainability. This approach allows us to understand the qualitative interaction of the three dimensions analyzed as a first step in understanding the implications of sustainability at the local level, taking into account its specificities, which are expressed in the data collected in the territory.

In this way, this approach makes it possible to show the impacts that the production of building materials generates simultaneously in the three dimensions mentioned above, without assigning weights, by directly expressing what the data show. The prioritization of one dimension over another will be a process that will have to be carried out when considering intervention processes to balance trade-offs or enhance synergies, and will depend on the context in which the intervention is to be carried out.

Finally, several questions arise from the synergies, trade-offs, and impacts identified: How to improve working conditions within artisanal production? What strategies could be used to achieve this without increasing the environmental impact and maintaining the use and circulation of the knowledge generated in the territory by artisanal producers? In other words, how do we convert trade-offs into synergies, in order to develop a production process that takes advantage of the potential of artisanal production and semi-mechanized production? One that, at the very least, makes it possible to achieve a neutral performance in most social impact subcategories, significantly reducing the impact on the social dimension. Additionally, there is a need to investigate production processes that can integrate the techniques, knowledge, and skills developed by artisanal producers with technological innovations and that do not result in a significant increase in environmental impacts.

It is certain that finding a balance between the dimensions of sustainability that were analyzed in this research implies, once again, the use of an interdisciplinary approach, so as not to fall into simplifications that lead us to prioritize one dimension over another, generating the imbalances that we have evidenced, thus enabling the design of strategies that allow the development of a sustainable construction sector.

The practical implications of this are equally complex. They require the construction sector to make an effort and challenge itself to better understand the context in which it is intervening; to understand building materials not only as finished products but as the result of a process that generates multiple impacts and how to integrate environmental, social, and cultural data in the decision-making of the whole construction sector.

Other practical implications can be developed and have positive impacts on the construction sector but must be carefully applied and monitored to avoid increasing the trade-offs identified. For example, developing public policies that encourage the use of traditional materials with lower environmental impacts, as well as establishing incentives to improve working conditions in artisanal production without considerably increasing its environmental impact. In terms of innovation in the production of materials, the development of hybrid technologies that combine the advantages of artisanal and semi-

mechanized production is required, integrating artisanal knowledge into semi-mechanized processes, without compromising cultural identity.

Other research that puts forward practical key principles that should guide strategies, solutions or policies to achieve a sustainable building sector places a strong emphasis only on the environmental, economic dimension and health impact that materials should have [51]. The main focus of this research is on the benefits of using locally produced sustainable materials, which on an environmental level reduce energy consumption and raw material use and on an economic level impact the local economies. This leaves out key dimensions, such as the social and cultural, which as presented in the previous sections of this document, play a key role in the sustainability of the production processes of local materials, even more so if we are talking about artisanal materials.

Efforts have been made to assess building material sustainability using Life Cycle Sustainability Assessment, which integrates environmental, social, and economic dimensions [52,53]. While material selection considers these dimensions, the social aspect remains a challenge, often limited to the impact of the buildings in the local community. Proper integration between dimensions is crucial. The cultural dimension has been even less addressed, as construction increasingly shifts from artisanal to technical processes.

In this context, it is complex to generate public policies that propose actions or regulations to achieve a construction sector that considers the three dimensions that this research proposes, let alone policies that assume the interactions between dimensions and the real complexity of sustainability at the local level. As in the aforementioned studies, this research raises certain guidelines that should be taken into account when generating public policies in favor of sustainability, and at an exploratory level tried to contribute to narrowing the knowledge gap related to the interaction between the three dimensions of sustainability that were analyzed.

Guidelines that will allow a sustainable construction sector in which the selection of construction materials is not only based on economic criteria of cost, market availability, and aesthetics, but also includes the cultural, social and environmental dimensions of sustainability, promoting the use of materials with less environmental impact, which contribute to improve the working conditions in the production process, respond effectively to the housing needs of communities with cultural pertinence and that do not affect the continuity of the producers' identity.

5. Conclusions

The main implications of the results obtained are directly related to the complexity of sustainability when approached from an interdisciplinary perspective. As a result, the factors that need to be considered to achieve a sustainable construction sector extend beyond the environmental dimension to include the social and cultural dimensions. This creates new challenges and opportunities. For example, the creation of public policies that can improve the social and working conditions in the construction sector is a challenge for a sector that has historically been characterized by its informality in Ecuador; linking the construction sector with policies for the preservation of heritage and cultural identity, not only at the architectural level, but above all thinking about the processes of production of building materials. This also presents certain challenges, since the construction sector has traditionally been a technical sector that only prioritized certain dimensions of sustainability.

In the same way, the results open the challenge of creating hybrid technologies that combine the advantages of artisanal and semi-mechanized production, ensuring the integration of traditional production in semi-mechanized processes, while preserving cultural identity.

Beyond these challenges, at the local level we were able to identify materials that open up new opportunities for the construction sector. Materials such as adobe, panel bricks, and bamboo have great potential for sustainability if the right public policies can be implemented to enhance the synergies identified.

In order to take advantage of these opportunities and to be able to solve these challenges, a process of coordination between the three key sectors that are involved in the construction sector at the national level is necessary. The public sector, academia, and the private sector have a key role to play, in order to set the necessary public policies to provide adequate practices to help achieve a sustainable construction sector, identify more materials with a high potential for sustainability, and develop new lines of research to solve the challenges of transitioning to a more sustainable construction sector.

Future research in this field can, thus, be directed in several directions: Firstly, generating a model for the selection of building materials that addresses sustainability from an interdisciplinary and life-cycle approach, and that simultaneously considers the environmental, social, and cultural impacts of building materials. The development of traditional materials with modern techniques to improve their social performance, without compromising their sustainability. On the other hand, it is essential to analyze the socio-economic impact of the transformation of production processes. Investigating how the replacement of an artisanal production process by a semi-mechanized one affects the local economy and the working conditions of the producers.

Emphasis should be placed on the importance of an integrated approach to sustainability in the construction sector, one that acknowledges the interplay between environmental, social, and cultural dimensions. Moving forward, it is crucial to foster collaboration between policymakers, academia, and industry stakeholders, to ensure that sustainability efforts translate into tangible improvements in working conditions, material innovation, and heritage preservation. By embracing interdisciplinary research and technological advancements, the construction sector in Ecuador can transition towards a more sustainable and socially responsible model. Ultimately, the successful implementation of these strategies will not only contribute to the sector's long-term viability but also promote inclusive economic growth and the conservation of cultural identity.

This research has certain limitations, mainly because of the exploratory nature of the proposed relations between the environmental, social, and cultural dimensions, which is why it does not present a quantitative analysis of the interaction between the three dimensions, which could be explored in future research. Another limitation is expressed in the findings, which were based on local data and a qualitatively representative sample, and provide a deep and contextualized understanding but are not statistically generalizable. Additionally, the exclusion of the economic dimension limited the scope, as it omits one of the three classical pillars of sustainability. However, this research identifies key elements within each of the three dimensions analyzed and proposes a qualitative interaction framework, offering a foundation for future interdisciplinary studies on the sustainability of building materials. In this regard, this study serves as a starting point for further investigations that can integrate broader sustainability dimensions and methodological approaches.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/su17051775/s1>, File S1: Production processes of selected materials; File S2: Technical data sheets of the materials; File S3: SLCA Survey Form—Building Materials Producers; File S4: Semi-structured Interview Script—Building Materials Producers. References [54–60] are cited in the supplementary materials.

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9. Kim, T.; Tae, S.; Chae, C.U. Analysis of Environmental Impact for Concrete Using LCA by Varying the Recycling Components, the Compressive Strength and the Admixture Material Mixing. *Sustainability* **2016**, *8*, 389. [CrossRef]
10. General Assembly of the United Nations. *Transforming Our World: The 2030 Agenda for Sustainable Development*; United Nations: New York, NY, USA, 2015.
11. Fernandes, J.; Peixoto, M.; Mateus, R.; Gervasio, H. Life cycle analysis of environmental impacts of earthen materials in the Portuguese context: Rammed earth and compressed earth blocks. *J. Clean. Prod.* **2019**, *241*, 118286. [CrossRef]
12. Ventura, A.; Ouellet-Plamondon, C.; Röck, M.; Hecht, T.; Roy, V.; Higuera, P.; Lecompte, T.; Faria, P.; Hamard, E.; Morel, J.-C.; et al. Environmental Potential of Earth-Based Building Materials: Key Facts and Issues from a Life Cycle Assessment Perspective. In *Testing and Characterisation of Earth-Based Building Materials and Elements*; Springer: Berlin/Heidelberg, Germany, 2021; pp. 261–296, ISBN 978-3-030-83297-1.
13. Heringer, A.; Howe, L.B.; Rauch, M. *Upscaling Earth: Material, Process, Catalyst*, 2nd ed.; GTA Verlag: Zurich, Switzerland, 2022; ISBN 978-3-85676-393-0.
14. Zea Escamilla, E.; Habert, G.; Lopez Muñoz, L.F. Environmental savings potential from the use of bahareque (mortar cement plastered bamboo) in Switzerland. In *Key Engineering Materials*; Trans Tech Publications Ltd.: Bäch, Switzerland, 2014; Volume 600, pp. 21–33. [CrossRef]
15. Escamilla, E.Z.; Archilla, H.; Nuramo, D.A.; Trujillo, D. *Bamboo: An Engineered Alternative for Buildings in the Global South. Bioclimatic Architecture in Warm Climates*; Springer: Berlin/Heidelberg, Germany, 2019; pp. 397–414. [CrossRef]
16. Arduin, D.; Caldas, L.R.; Paiva, R.d.L.M.; Rocha, F. Life Cycle Assessment (LCA) in Earth Construction: A Systematic Literature Review Considering Five Construction Techniques. *Sustainability* **2022**, *14*, 13228. [CrossRef]
17. Ecoinvent—Data with Purpose. Available online: <https://ecoinvent.org/> (accessed on 3 September 2024).
18. Martínez-Rocamora, A.; Solís-Guzmán, J.; Marrero, M. LCA databases focused on construction materials: A review. *Renew. Sustain. Energy Rev.* **2016**, *58*, 565–573. [CrossRef]
19. INEC—Encuesta Nacional de Empleo, Desempleo y Subempleo. Available online: <https://www.ecuadrencifras.gob.ec/empleo-desempleo-y-subempleo/> (accessed on 10 September 2023).
20. Abowitz, D.A.; Toole, T.M. Mixed method research: Fundamental issues of design, validity, and reliability in construction research. *J. Constr. Eng. Manag.* **2010**, *136*, 108–116. [CrossRef]
21. Escamilla Zea, E.; Habert, G.; Wohlmuth, E. When CO₂ counts: Sustainability assessment of industrialized bamboo as an alternative for social housing programs in the Philippines. *Build. Environ.* **2016**, *103*, 44–53. [CrossRef]
22. Hermida, A.; Cabrera, N.; Molina, L. *Casas Y Conjuntos Vivienda social en Cuenca Entre 1973 y 2014*; UCuenca Press: Cuenca, Ecuador, 2021; ISBN 978-9978-14-464-0.
23. Hays, K.M. Critical Architecture Between Culture and Form. *Perspecta* **1984**, *21*, 14–29. [CrossRef]
24. Luukkanen, J.; Vehmas, J.; Panula-Ontto, J.; Allievi, F.; Kaivo-oja, J.; Pasanen, T.; Auffermann, B. Synergies or Trade-Offs? A New Method to Quantify Synergy Between Different Dimensions of Sustainability. *Environ. Policy Gov.* **2012**, *22*, 337–349. [CrossRef]
25. Zhong, R.; Pei, F.; Yang, K.; Xia, Y.; Wang, H.; Yan, G. Coordinating socio-economic and environmental dimensions to evaluate regional sustainability—Towards an integrative framework. *Ecol. Indic.* **2021**, *130*, 108085. [CrossRef]
26. National Academy of Sciences. *Facilitating Interdisciplinary Research*; The National Academies Press: Washington, DC, USA, 2005. [CrossRef]
27. Liu, J.; Mooney, H.; Hull, V.; Davis, S.J.; Gaskell, J.; Hertel, T.; Lubchenco, J.; Seto, K.C.; Gleick, P.; Kremen, C.; et al. Systems integration for global sustainability. *Science* **2015**, *347*, 963–974. [CrossRef]
28. Roy, B. *Multicriteria Methodology for Decision Aiding*; Springer-Science+Business Media: Berlin/Heidelberg, Germany, 1996. [CrossRef]
29. INEC. *Boletín Técnico N°01-2021-ENED Encuesta Nacional de Edificaciones (ENED)*, 2020; Dirección de Estadísticas Económicas: Quito, Ecuador, 2021.
30. Fink Hafner, D.; Dagen, T.; Dousak, M.; Novak, M.; Hafner-Fink, M. Delphi Method: Strengths and Weaknesses. *Metod. Zv.* **2019**, *16*, 1–19. [CrossRef]
31. INBAR. Evaluación Regional De Recursos De Bambú Mediante La Aplicación Móvil Provincia de Manabí, Ecuador. In *Documento de trabajo del INBAR*, 82nd ed.; INBAR: Beijing, China, 2019.
32. INPC. *Arquitectura Tradicional en Azuay y Cañar Técnicas, Creencias, Prácticas y Saberes*; INPC: Cuenca, Ecuador, 2011.
33. Aguirre Ullauri, M.C.; Romero Bastidas, M.; Ramírez Bustamante, J.; Castillo Carchipulla, E. El ladrillo artesanal de Cuenca (Ecuador): Una caracterización inicial en el marco patrimonial. *Conserv. Patrimonio* **2022**, *41*, 68–94. [CrossRef]
34. García Vélez, G.; Amaya Ruiz, J. *De Cadena de Producción Hacia Cadena de Valor del Adobe: Patrimonio Como Recurso de Desarrollo*; Universidad de Cuenca: Cuenca, Ecuador, 2024; ISBN 978-9978-14-539-5.
35. ISO 14044:2006(en); Environmental Management—Life Cycle Assessment—Requirements and Guidelines. ISO: Geneva, Switzerland, 2006. Available online: <https://www.iso.org/obp/ui/#iso:std:iso:14044:ed-1:v1:en> (accessed on 30 January 2023).

36. UNEP. *Guidelines for Social Life Cycle Assessment of Products and Organizations*; United Nations Environment Programme (UNEP): Nairobi, Kenya, 2020.
37. Ecoinvent. *The Ecoinvent Database. A Life Cycle Inventory for Transparency in Environmental Assessments*; Ecoinvent: Zurich, Switzerland, 2022. Available online: <https://support.ecoinvent.org/ecoinvent-version-3.9.1> (accessed on 10 January 2023).
38. Ecoinvent Glossary. Available online: <https://Ecoinvent.org/glossary-terms/> (accessed on 10 January 2023).
39. Moreno Ruiz, E.; Lérová, T.; Valsasina, L. *Guidelines for Data Providers to the Ecoinvent Database*; Ecoinvent: Zurich, Switzerland, 2016.
40. The Intergovernmental Panel on Climate Change (IPCC). *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*; The Intergovernmental Panel on Climate Change (IPCC): Geneva, Switzerland, 2019; ISBN 978-4-88788-232-4.
41. Zampori, L.; Pant, R. *Suggestions for Updating the Product Environmental Footprint (PEF) Method*; EUR 29682 EN; Publications Office of the European Union: Luxembourg, 2019; ISBN 978-92-76-00653-4.
42. EC-JRC European Platform on Life Cycle Assessment—Environmental Footprint. Available online: <https://eplca.jrc.ec.europa.eu/EnvironmentalFootprint.html> (accessed on 7 January 2023).
43. IPCC Climate Change 2013. *The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; Stocker, T.F., Qin, D., Plattner, G.-K., Tignor, M., Allen, S.K., Boschung, J., Nauels, A., Xia, Y., Bex, V., Midgley, P.M., Eds.; Cambridge University Press: Cambridge, UK; New York, NY, USA, 2013.
44. UNEP. *Methodological Sheets for Subcategories in Social Life Cycle Assessment (S-LCA)*; United Nations Environment Programme (UNEP): Nairobi, Kenya, 2021.
45. Guber, R. *La Etnografía: Método, Campo Y Reflexividad*; Grupo Editorial Norma: Cali, Colombia, 2001.
46. Foucault, M. *El orden del Discurso*; Tusquets Editores: Buenos Aires, Argentina, 2005.
47. Dierckx de Casterle, B.; Gastmans, C.; Bryon, E.; Denier, Y. QUAGOL: A guide for qualitative data analysis. *Int. J. Nurs. Stud.* **2011**, *49*, 360–371. [[CrossRef](#)] [[PubMed](#)]
48. Thomas, D. A General Inductive Approach for Analyzing Qualitative Evaluation Data. *Am. J. Eval.* **2006**, *27*, 237–246. [[CrossRef](#)]
49. Hossain, U.; Poon, C.S.; Dong, Y.H.; Lo IM, C.; Cheng JC, P. Development of social sustainability assessment method and a comparative case study on assessing recycled construction materials. *Int. J. Life Cycle Assess* **2018**, *23*, 1654–1674. [[CrossRef](#)]
50. Mahamadu, A.-M.; Baffour Awuah, K.; Booth, C.A. Principles of sustainability and life-cycle analysis. In *Sustainability of Construction Materials*; Elsevier Ltd.: Amsterdam, The Netherlands, 2016; pp. 13–31.
51. Yahia, A.K.M.; Shahjalal, M. Sustainable Materials Selection in Building Design and Construction. *Int. J. Sci. Eng.* **2024**, *1*, 106–119. [[CrossRef](#)]
52. Tran, D.B.; Tran, V.T.; Pham, X.A.; Nguyen, V.T. A General Framework for Sustainability Assessment of Buildings: A Life-Cycle Thinking Approach. *Sustainability* **2023**, *15*, 10770. [[CrossRef](#)]
53. Figueiredo, K.; Pierott, R.; Hammad, A.W.A.; Haddad, A. Sustainable material choice for construction projects: A Life Cycle Sustainability Assessment framework based on BIM and Fuzzy-AHP. *Build. Environ.* **2021**, *196*, 107805. [[CrossRef](#)]
54. Ministerio De Vivienda, Construcción Y Saneamiento. *Norma E.080 Diseño Y Construcción Con Tierra Reforzad*; Ministerio De Vivienda, Construcción Y Saneamiento: Lima, Peru, 2017.
55. Ministerio de Desarrollo Urbano y Vivienda. *Norma Ecuatoriana de la Construcción Eficiencia Energética en Edificaciones Residenciales (EE)*; Ministerio de Desarrollo Urbano y Vivienda: Quito, Ecuador, 2018.
56. Servicio ecuatoriano de normalización (INEN). *NTE INEN 3049. Ladrillos cerámicos. Parte 2: Unidad de mampostería estructural. Requisitos*; Servicio ecuatoriano de normalización (INEN): Quito, Ecuador, 2019.
57. Servicio ecuatoriano de normalización (INEN). *NTE INEN 3049. Ladrillos cerámicos. Parte 3: Unidad de mampostería estructural. Requisitos*; Servicio ecuatoriano de normalización (INEN): Quito, Ecuador, 2019.
58. Ministerio de Desarrollo Urbano y Vivienda. *Norma ecuatoriana de la construcción—Estructuras de guadúa (GaK)*; Ministerio de Desarrollo Urbano y Vivienda: Quito, Ecuador, 2016.
59. Servicio ecuatoriano de normalización (INEN). *INEN 990: Tejas ceramicas. Requisitos*; Servicio ecuatoriano de normalización (INEN): Quito, Ecuador, 1982.
60. Servicio ecuatoriano de normalización (INEN). *NTE INEN 3066. Bloques de hormigón. Requisitos y métodos de ensayo*; Servicio ecuatoriano de normalización (INEN): Quito, Ecuador, 2016.

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